

DIAN RAISA GUMILAR

Informatics Engineering | AI & Machine Learning | Computer Networks

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PROFESSIONAL SUMMARY

Informatics graduate from Institut Teknologi Nasional (Itenas) Bandung specializing in Artificial Intelligence and Machine Learning. Proficient in Python-based data processing, Natural Language Processing (NLP), and predictive model evaluation. Demonstrates strong analytical thinking, systematic problem-solving, and cross-functional collaboration, eager to drive impactful software and AI solutions.

WORK EXPERIENCE

Machine Learning Laboratory Assistant 02/2026 – 06/2026
Institut Teknologi Nasional Bandung Neglasari, Cibeunying Kaler, Kota Bandung, West Java
Led hands-on data science and machine learning laboratory sessions across 10 structured modules for 50+ undergraduate students, utilizing Python (Pandas, Matplotlib, Scikit-Learn) to demonstrate end-to-end data preprocessing, exploratory analysis, and classification/clustering algorithms. Provided technical debugging support for the cohort and direct mentorship to 8 students, guiding them to top academic results.

Computer Networks Teaching Assistant 04/2026 – 05/2026
Institut Teknologi Nasional Bandung Neglasari, Cibeunying Kaler, Kota Bandung, West Java
Stepped in as a substitute teaching assistant to ensure uninterrupted learning continuity for the Computer Networks course, leading hands-on lab sessions on VLAN segmentation, inter-VLAN routing, and OSPF dynamic routing. Utilized eNSP network simulation tools and CLI environments to conduct live configuration demonstrations, troubleshoot setup errors, and assist students with practical debugging

Research & Development / IT Internship 03/2024 – 06/2024
National Research and Innovation Agency (BRIN) Dago, Coblong, Kota Bandung, West Java
Engineered an NLP and BERTopic text-processing pipeline in Python (Pandas, NumPy) to develop "Collab Identifier," a research recommendation prototype analyzing cross-institutional collaboration patterns—resulting in an officially registered software copyright (HKI - DJKI) and an international journal publication in *Emerald Publishing (Library Hi Tech)*.

EDUCATION

Bachelor of Informatics 2022 – 2026
Institut Teknologi Nasional Bandung (Itenas) Bandung, West Java, Indonesia
Cumulative GPA: 3.58 / 4.00 Specialization: Artificial Intelligence, Machine Learning, Deep Learning, and Computer Networks.

TRAINING & CERTIFICATIONS

Huawei ICT Associate (HCIA) AI V3.5
Huawei Technologies | Aug 2024 – Aug 2027
Score: 960 / 1000
Key Learnings: Machine Learning and Deep Learning architectures, Python predictive modeling, MindSpore framework, ResNet-50 image classification, and end-to-end cloud training pipelines using ModelArts.
Impact & Outcome: Achieved an outstanding score of 960/1000, demonstrating certified readiness to develop, evaluate, and deploy scalable computer vision and AI solutions in enterprise production environments.

Basics of Network Layer Protocols
Huawei Technologies | Dec 2025 – Dec 2027
Key Learnings: Fundamental network infrastructure concepts, IPv4/IPv6 addressing schemes, packet forwarding mechanisms, and routing protocol operation.
Impact & Outcome: Built a robust technical foundation for diagnosing packet-level network bottlenecks, troubleshooting routing anomalies, and managing complex multi-subnet network environments.

PUBLICATIONS & INTELLECTUAL PROPERTY

Collab Identifier: A Prototype for Uncovering Thematic Research Collaboration Opportunities 10/2026
Emerald Publishing — Library Hi Tech (Indexed International Journal)
Authored the peer-reviewed research paper introducing an NLP- and BERTopic-driven framework to uncover latent research domains and potential cross-institutional collaborations.

Integrating BERTopic and Heterogeneous Knowledge Graphs for Research Recommendation 10/2026
IEEE — The 13th International Conference on Computer, Control, Informatics, and its Applications (IC3INA)
Co-authored and presented research on designing a temporal collaboration recommendation architecture combining BERTopic with Heterogeneous Knowledge Graphs.

Registered Software Copyright (HKI): Collab Identifier 06/2025
Directorate General of Intellectual Property (DJKI), Ministry of Law and Human Rights RI
Awarded official intellectual property protection for the computer software "Collab Identifier", an analytical platform bridging NLP and Library & Information Science (LIS) to evaluate cross-institutional research synergies.

HONORS & AWARDS

Graduated with Honors (Cum Laude) – Bachelor of Informatics 01/09/2026
Institut Teknologi Nasional Bandung

1st Place Winner – Huawei ICT Competition 2025–2026 (National Final, Computing Track) 12/2025
Huawei Technologies Indonesia
Outperformed national university teams across Indonesia by solving advanced networking and cloud-computing challenges, demonstrating strong technical execution and problem-solving under competition environments.

LEADERSHIP & STUDENT GOVERNANCE

Informatics Student Association (HMIF) Itenas Bandung 07/2024 – 12/2024
Staff of Internal Audit Unit
Monitored organizational governance, operational procedures, and program compliance across student association divisions to ensure operational transparency and accountability. Conducted performance evaluations and administrative audits on student work programs, delivering formal evaluation reports to leadership boards.

Informatics Student Association (HMIF) Itenas Bandung 09/2023 – 06/2024
Staff of Cadre Division
Co-designed and executed character-building curricula and orientation programs aimed at accelerating cultural adaptation for incoming informatics freshmen. Facilitated leadership development workshops, collaborative activities, and mentorship initiatives to nurture team synergy.

SKILLS

Programming & Data
Python, Dart, SQL, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Git

AI & Data Science
Machine Learning, Deep Learning, Natural Language Processing (NLP), Topic Modeling (BERTopic), Computer Vision, Knowledge Graphs, Model Training & Evaluation

Networking & Systems
Computer Networks, Network Layer Protocols, VLAN Configuration, OSPF, eNSP, Linux, IoT

Soft Skills & Languages:
Analytical Thinking, Systematic Problem Solving, Teamwork, Mentoring, Audit, Time Management

LANGUANGE

English
Professional Working

Indonesian
Native/Fluent